

Family 1.gf in two pages

the world’s ocean and air, measured finer than 10 km and 5 days

blauewelt/earth — two-page summary of the design note, 17 September 2026. A glossary of every term follows the two pages.

What it is. Family 1.gf is a proposed set of datasets (“stores”) that gives the programme’s forecasting model the whole globe’s ocean and atmosphere as measured at **10 km or finer, every 5 days or finer**: ship reports back to 1662, every non-Argo ocean profile since 1772, ship bottle chemistry, chemistry floats, open-ocean moorings, and satellite fields kept at their own resolution — ocean colour, sea-surface temperature (SST), sea ice, swath sea level, wave height, cloud-top temperature and column CO₂. **What it is for:** the programme’s four prediction goals — El Niño, the ocean carbon sink, the Atlantic overturning circulation (AMOC) and SST — each of which has a mechanism at these scales (wind bursts, fronts, blooms, the ice edge, boundary currents); every store must lower the model’s error on one of them or be dropped. **What it is not:** not a land dataset (that is family 1.0.tf); not a replacement for the coarser global tensor (family 7.2, 0.25°), which stays unchanged; and not gap-filled analyses — raw measurements first, a derived product only where it carries measurements nothing else does. Three sources slightly coarser than the rule are admitted as named exceptions (E1–E3). A source whose licence forbids republishing is used privately and never redistributed.

store	what it is	producer	space	time	record	stored size	track	phase
seaice_asi	daily sea-ice cover from microwave radiometers	Univ. Bremen (ASI, AMSR-E/AMSR2)	6.25 km polar grids (3.125 km exists)	daily	2002-06 → 2011-10; 2012-07 → now	≈ 30 GB	public if Bremen agrees, else private	A
oc4k	daily ocean colour: chlorophyll, water clarity, reflectance	ESA OC-CCI v6.0 (PML)	4.6 km (1/24°)	daily	1997-09 → 2026-06	0.3–0.4 TB; ≈ 1.1 TB with the 6 Rrs bands (B)	public	A
icoads	ship and buoy weather reports	NOAA NCEI / NCAR (ICOADS 3.0)	point	per report	1662 → 2026-08	note ≈ 75 GB. Probe (2010-07): 2.02 M reports, 47 B/row ⇒ ≈ 1.37 B rows, ≈ 64 GB	public	A
wod	ocean temperature, salinity and oxygen profiles, Argo excluded	NOAA NCEI (WOD 2023)	point profile	per cast	1772 → 2026	note 2 GB. Probe (2005-03): 40,541 casts, 287 B/row ⇒ ≈ 14.7 M casts, ≈ 4.2 GB	public	A
glodap	ship bottle samples of carbon, nutrients and oxygen at depth	GLODAPv3 (NOAA NCEI)	point bottle	per cruise	1972–2023	note 80 MB. Probe (whole archive): 1,460,215 bottles, 53 B/row, ≈ 77 MB	public	A
bgcarg	floats that also measure oxygen, nitrate, pH, chlorophyll	Argo GDAC (BGC-Argo)	point profile	≈ 10 days per float	2002 →	note 80 MB. Probe (2022-05): 1,717 profiles, 219 B/row ⇒ ≈ 74 MB (75 GB download)	public	A
oceansites	fixed open-ocean moorings	OceanSITES (NDBC copy)	point mooring	hourly or finer	1980s →	note ≈ 4 GB. Probe (2018-09): 344,798 rows from 17 sites, 83 B/row (29 MB a month)	public	A
xco2	CO ₂ averaged through the whole air column	NASA OCO-2, OCO-3 (GES DISC); GOSAT via ACOS	< 3 km ² sounding	per sounding (70–120 k a day)	2014-06 →; OCO-3 2019-08 →	≈ 18 GB	public	A
swot	sea-surface height across two 50 km radar swaths	NASA/CNES SWOT (PO.DAAC)	2 km	per pass (20.86-day cycle)	2023-07 →	≈ 0.6 TB	public	B
swh	wave height along altimeter tracks	ESA CCI Sea State (Copernicus Marine)	≈ 7 km along track	1 per second	1991-08 → 2023-12	≈ 70 GB	public	B
sst_cci05	daily SST per satellite sensor, merged by our builder	ESA SST CCI v3.0 (CEDA)	5.6 km (0.05°)	daily, day and night	1979-07 → 2024; SLSTR 2016-06 →	≈ 1.2 TB	public	B
irtb	cloud-top temperature from geostationary weather satellites	NOAA CPC (NASA GES DISC)	4 km	30 min (3-hourly in phase B)	1998-01 → now	≈ 0.7 TB tropics 3-hourly; ≈ 8 TB full (C)	public	B / C
precip01 (E2)	daily satellite rainfall (IMERG)	NASA/JAXA GPM	11 km (0.1°)	daily	10,100 days to 2025-09 (V07 Final); V08 expected	0.2–0.3 TB	public	B
deep_arrays	instrument records of the four Atlantic overturning arrays	RAPID/MOCHA (BODC), OSNAP (Duke), SAMBA (AOML), MOVE	point mooring	15 min–1 h [†]	2000 / 2004 / 2009 / 2014 →	< 5 GB	public (RAPID terms to confirm)	B
sst_acspo02	daily SST merged across sensors	NOAA STAR ACSPO (PO.DAAC, CoastWatch)	2.2 km (0.02°)	daily (AM and PM)	2000-02 → now –2 months	≈ 2.3 TB	public	C
wind12 (E1)	ocean wind from scatterometers	EUMETSAT OSI SAF (ASCAT); QuikSCAT	12.5 km swath cells	per pass	ASCAT 2007 →; QuikSCAT 1999–2009	≈ 1.7 TB (or ≈ 50 GB as five-day 0.125° frames)	public	C

Sizes are the note’s estimates unless marked **Probe**: a one-month test fetch through the real code (E-082, 17 Sep 2026), giving rows and stored bytes per row, and, where stated, the whole store scaled from it. [†] estimate not verifiable at the source. B = bytes. Already held elsewhere and not rebuilt: OISST, NCEP, GLORYS-derived and ocean-colour grids (family 7.2); Argo (family 8); drifters, tropical moorings, ship CO₂, along-track sea level (family 10.1); fishing effort (family 10.2).

How it is stored. One registry, `tensors/family1_gf/family1gf.json` on the project’s public Hugging Face repository, lists every store with its tier, location, channels, checksums, licence and track. *Point stores* (tier P) are nine column arrays sorted by five-day bin, with an index of where each bin starts; a record older than 1914 gets a 64-bit time column (schema 3). *Fine grids* (tier G-fine) are one compressed file per five-day bin, holding one frame per day (or per three hours), each cut into 256×256 -pixel tiles with an index of byte positions, so a read fetches only the few tiles around the place being predicted and an empty tile costs nothing. Every value reaches the model as a token carrying its own footprint — the area and time span it averages — so nothing coarse is copied into fine cells. Private stores use the same format in a private repository and are listed in the public registry. Credentials stay on GitHub’s build machines; rented machines hold none.

The ladder

- A** 8 stores, ≈ 0.5 TB, no account or a hosted runner’s login, under \$10: `seaice_asi`, `oc4k`, `icoads`, `wod`, `glodap`, `bgcargo`, `oceansites`, `xco2`; the registry. Five of the point stores have run their one-month probes.
- B** the terabyte stores, ≈ 3.5 TB, \$20–40: `swot`, `sst_cci05`, `irtb` (tropics, 3-hourly), `precip01`, `swh`, `deep_arrays`, and `oc4k`’s six reflectance bands.
- C** decisions once A and B have verdicts: `sst_acspo02`, `wind12`, the full half-hourly `irtb`, PACE colour into `oc4k`.

Glossary

Every term, acronym, producer, instrument and code name used in this summary and in the full design note, in alphabetical order.

1.gf channel names — `sst`, `airt`, `slp`, `wind_u`, `wind_v`, `dewpt`, `wave_h`, `cloud`: ship-report SST, air temperature, sea-level pressure, east and north wind, dew point, wave height, cloud cover; `chlora`, `kd_490`, `nobs`, `log_chl`: chlorophyll, light attenuation at 490nm, observation count, log-chlorophyll; `sic`: sea-ice concentration; `sshakar`, `sig0`, `quality`: SWOT height anomaly, radar backscatter, quality flag; `sw_h`, `sw_h_denoised`, `sw_h_unc`: wave height, smoothed, and its uncertainty; `dtime`: time offset of a pixel within its pass; `tb`: brightness temperature; `precip`: precipitation; `u10`, `v10`: 10m wind components; `xco2`, `xco2_unc`, `surface`: column CO₂, its uncertainty, land or water; `depth`, `bpr`: instrument depth, bottom pressure; DOXY, CHLA, BBP700, NITRATE, PH: float oxygen, chlorophyll, particle backscatter, nitrate, pH.

ACOS — NASA’s retrieval of column CO₂ applied to GOSAT’s measurements.

ACSPO — NOAA’s Advanced Clear-Sky Processor for Ocean, which turns satellite radiances into SST.

ALOHA — A cabled sea-floor observatory north of Hawaii; its 162 GB record is left out of `oceansites`.

AlphaEarth Foundations (AEF) — Google DeepMind’s model that turns sequences of satellite images into one 64-number description of each 10m of land for each year; its published output is the *Satellite Embedding*.

altimeter, along-track — A radar satellite that measures sea-surface height straight below it; along-track data are its measurements along that ground line.

AMOC — The Atlantic Meridional Overturning Circulation: the system of Atlantic currents that carries warm water north near the surface and cold water south at depth, one of the programme’s four prediction goals.

AMSR-E, AMSR2, AMSR3 — Japanese microwave radiometers (2002–2011, 2012 on, 2025 on) that see sea ice and SST through cloud; AMSR-Unified is NSIDC’s merged product line (`AU_SI6`, `AU_SI12`).

anchor — The place and time the model is asked to predict; everything it reads is placed relative to it.

annual target — A yearly map the model is asked to predict (a loss year, a biomass map); it is filed at the end of its year so it is never read as an input from inside that year.

AOML — NOAA’s Atlantic Oceanographic and Meteorological Laboratory, Miami.

Argo, BGC-Argo, Sprof — The global array of about 4,000 drifting floats that profile temperature and salinity to 2,000 m; BGC-Argo floats add chemistry sensors; Sprof files merge each float’s variables.

ARTIST, ASI — The ARTIST Sea Ice algorithm of the University of Bremen, which computes ice concentration from 89 GHz microwave channels.

ASCAT — The Advanced Scatterometer on Europe’s MetOp satellites.

attribution, yes, no (licence.redistribution) — Registry values saying whether a licence lets the data be republished freely, with credit, or not at all.

AVHRR — The Advanced Very High Resolution Radiometer, an infrared imager on NOAA and MetOp weather satellites since 1978.

bgcargo — Family 1.gf store of BGC-Argo float profiles of oxygen, nitrate, pH, chlorophyll and backscatter at 16 depths.

bin_offsets, CSR — The index that says where each five-day bin’s rows start in a point store (compressed sparse row).

bin (pentad) — The five-day time step every family uses, counted from 1 January 1982; a bin before 1982 has a negative number.

bin_rule, epoch, pentad_days — Registry fields that fix how a time becomes a bin: the start date 1982-01-01 and the five-day step.

BODC — The British Oceanographic Data Centre.

box (rented, verified) — A machine hired by the hour from a marketplace; it holds no credentials, and a verified one runs on a host the marketplace has vetted.

brightness temperature — The temperature a satellite infers from the radiation it receives; over cloud it is the cloud top’s temperature.

B/row (bytes per row) — Stored size of one row of a point store; a row costs 27 + 2 bytes per channel (31 + 2 per channel with the 64-bit time).

CC0, CC BY, CC BY-NC — Creative Commons licences: public domain; free with attribution; free with attribution for non-commercial use only.

C (channels) — The number of values stored per row or per pixel.

CDR — Climate data record: a long series processed consistently for climate use.

CDS — The Copernicus Climate Data Store (account required).

CEDA — The UK Centre for Environmental Data Analysis, an open archive.

CFCs (CFC-11, CFC-12) — Chlorofluorocarbons, man-made gases whose dissolved amount dates when water last touched the surface.

checksum (sha256) — A 64-character fingerprint of a file; if one byte changes, the fingerprint changes.

chfrank/earth-tensors, chfrank/earth-tensors-private — The project’s public and private Hugging Face dataset repositories.

climatology — The long-term average for a place and season — the simplest forecast a model has to beat.

CNES — France’s space agency.

CoastWatch — NOAA’s programme that serves satellite ocean data.

cone — The set of past measurements the model reads around the anchor: wider for older times, like a cone in space and time.

Copernicus, Copernicus Marine — The EU’s Earth-observation programme and its marine service (free account required).

CORA — Copernicus’ quality-controlled collection of temperature and salinity profiles.

CPC — NOAA’s Climate Prediction Center.

CSV — Comma-separated values, a plain-text table format.

DAAC (LP, ORNL, PO, OB) — NASA’s discipline data centres: LP DAAC for land, ORNL DAAC for ecology, PO.DAAC for oceans, OB.DAAC for ocean biology.

DAC, GDAC — Data assembly centre, and the global one that gathers a network’s data (Argo, gliders, OceanSITES).

deep_arrays — Family 1.gf store of instrument records from the four Atlantic overturning arrays.

degraded — Registry list of every check a build had to relax, so a relaxation can never be silent.

derived_works — Registry field saying whether a trained model may be published under the source’s licence: **free, non-commercial** or **restricted**.

DIC, TA (alkalinity) — Dissolved inorganic carbon and total alkalinity, the two numbers that fix seawater’s carbon chemistry.

distribution — Registry field naming a store’s track, **public** or **private**.

done.json — A marker a build writes last, so an unfinished build can never look finished.

DPR, GMI — GPM’s Dual-frequency Precipitation Radar and its microwave imager.

dtype; int8, int16, int32, int64, uint8, float16, float32 — The numeric type values are stored in: whole numbers of 1, 2, 4 or 8 bytes (uint8 unsigned), or decimals of 2 or 4 bytes.

DUACS — The CNES and Copernicus system that turns altimetry into gridded sea-level maps.

Duke — Duke University, which hosts OSNAP’s data.

E-077 — The experiment that added ocean colour to the global tensor (family 7.1).

E-078, E-078c — The design study that gave family 10 one schema for every granularity, and its first experiment training a local image codec on about 300 coastal cells.

E-079 — Family 10’s first data pull, which built the four family-10.1 point stores.

E-081 — The experiment that built family 10.2’s fishing-effort store and its globe layer.

E-082 — The build plan for families 1.gf, 1.0.tf and 0.9.tf; its first wave wrote and probed the phase-A point-store adapters.

E1–E6 (exceptions) — Sources admitted although they break the 10 km / 5-day rule: E1 scatterometer wind at 12.5 km, E2 IMERG rain at 11 km, E3 GOSAT at 10.5 km, E4 CHIRPS rain, E5 PALSAR yearly mosaics, E6 sub-metre photos.

Earthdata login — NASA’s free user account, needed to download most NASA data.

embedding — A learned list of numbers summarising what was observed at a place, arranged so that similar places get similar numbers.

ERDDAP — A data server that answers subset queries over HTTP.

ERSST — NOAA’s Extended Reconstructed SST, a monthly analysis back to 1854.

ESA — The European Space Agency.

ESA CCI — ESA’s Climate Change Initiative, which builds long consistent climate records from satellites.

EUMETSAT — Europe’s weather-satellite agency.

falsifier — The test, fixed before building, that can show a store is not worth keeping.

family — A numbered set of datasets built to one storage design and read by the programme’s forecasting model.

family 0.9.tf — Family 1.0.tf with its raw-imagery catalogue replaced by Google’s AlphaEarth embedding, everything else inherited.

family 10, 10.1, 10.2 — The granularity-aware dataset set in which every measurement keeps its own place, time and footprint; 10.1 built four point stores (drifters, tropical moorings, ship CO₂, satellite sea level) and 10.2 inherited them and added fishing effort.

family10.json, family1gf.json, family1tf.json, family09tf.json — The registry files of families 10, 1.gf, 1.0.tf and 0.9.tf.

family 1.0.tf — The proposed land-and-coast family at the same scale, built from raw observations and a catalogue of raw imagery (“tf”: terrestrial, fine).

family 1.gf — The proposed global family of ocean and atmosphere measurements finer than 10 km and 5 days (“gf”: global, fine).

family 7, 7.1, 7.2 — The published global input tensor: every 0.25° cell from pole to pole in five-day steps, 1982–2024; 7.1 added ocean colour and 7.2 is the current version.

family 8 — The programme’s store of Argo float temperature and salinity profiles.

fishing — Family 10.2’s store of apparent fishing effort per vessel from Global Fishing Watch.

footprint; log₂ f_p, log₂ f_t (log2_fp, log2_dt, fp) — The area and time span a value averages over, stored as base-2 logarithms relative to 27.83 km and 5 days; −4 is the label for a point measurement with no area.

forcing — An input that drives the system being predicted, such as wind over the ocean or rain over land.

foundation temperature — The temperature of the water just below the layer the sun warms each day.

frame; frames_per_bin, frame_seconds, F — One time slice of a gridded field; a five-day bin holds F of them (5 for daily, 40 for three-hourly), each lasting **frame_seconds**.

FTP — File Transfer Protocol, an old way of serving files.

g025, g100, rg100, oc025 — Family 7.2’s four gridded groups: 0.25° ocean surface, 1° atmosphere and land, 1° monthly ocean interior, 0.25° ocean colour.

GDEX — NCAR’s Geoscience Data Exchange archive.

GDP — NOAA’s Global Drifter Program of free-drifting surface buoys.

geostationary — Orbiting at the Earth’s rotation rate, so a satellite watches the same half of the globe continuously.

GES DISC — NASA’s Goddard Earth Sciences data centre, for atmosphere and precipitation data.

GHz — Gigahertz, the unit of a microwave channel’s frequency.

GitHub, hosted runner — The code-hosting service the project uses; a hosted runner is one of GitHub’s own build machines, the only place credentials are used.

glodap — Family 1.gf store of GLODAP ship bottle samples.

GLODAP, GLODAPv3 — The Global Ocean Data Analysis Project’s quality-controlled synthesis of ship bottle chemistry, version 3 (July 2026).

GLORYS — Copernicus’ global ocean reanalysis.

GOSAT — Japan’s Greenhouse gases Observing SATellite.

GPM, GPM_MERGIR — NASA and JAXA’s Global Precipitation Measurement mission; MERGIR is the merged geostationary infrared product served beside it.

gtmba — Family 10.1’s store of the tropical moored buoy arrays, including TAO.

HTTP, HTTPS, HTTP range — The web’s transfer protocol (HTTPS is the encrypted form); a range request asks for one byte span of a file.

Hub (Hugging Face) — Hugging Face’s dataset hosting, where the project publishes its stores; it limits files per repository and per folder.

icoads — Family 1.gf store of ICOADS ship and buoy weather reports since 1662.

ICOADS, IMMA1 — The International Comprehensive Ocean-Atmosphere Data Set of ship and buoy reports since 1662, and its record format.

Ifremer — France’s national ocean research institute.

IMERG — NASA’s merged satellite precipitation estimate (half-hourly and daily, 0.1°); V07 and V08 are versions, Final the gauge-corrected run.

irtb — Family 1.gf store of 4 km geostationary cloud-top (infrared brightness) temperature.

JAXA — Japan’s space agency.

JPL — NASA’s Jet Propulsion Laboratory.

JSON — A plain-text format for structured data, used for every registry.

KaRIn — SWOT’s Ka-band Radar Interferometer, which measures height across two 50 km swaths.

KB, MB, GB, TB, PB — Kilo-, mega-, giga-, tera- and petabytes: each a thousand times the last.

Kelvin wave — An equatorial ocean wave that carries a bump of sea level eastward across the Pacific in about two months, a precursor of El Niño.

lat, lon, values, qc, platform, time_s — Arrays of a point store: position, the measured values, a quality code, a hashed instrument identity, and the time in seconds since 1982.

layout, prefix, path, cadence, footprint, tier — Registry fields per group: storage layout, folder, the group’s own location, time step, footprint and tier.

LEO — Low Earth orbit, a few hundred kilometres up; the ACSPO L3S-LEO product merges such satellites.

level 1B, 2, 3, 3C, 3S, 4 (L1B, L2, L3, L3C, L3S, L4) — Processing stages of satellite data: calibrated signals; measurements on the instrument’s own swath; measurements binned onto a grid; binned per sensor; merged across sensors; and gap-filled analyses, which are derived.

masked-token loss — The model’s error when asked to predict tokens hidden from it; lower means the inputs helped.

MetOp — Europe’s polar-orbiting weather satellites (A, B, C).

ml/CLAUDE.md, ml/family10.store.py — The project’s rules for machine-learning work (security, replication), and the code that reads a store.

MOCHA, MOVE, OSNAP, RAPID, SAMBA — The four Atlantic overturning arrays: RAPID/MOCHA at 26.5°N, MOVE at 16°N, OSNAP in the subpolar North Atlantic, SAMBA at 34.5°S.

MODIS; Terra, Aqua — The Moderate Resolution Imaging Spectroradiometer on NASA’s Terra (1999) and Aqua (2002) satellites, both near the end of their lives.

MUR, OSTIA — Gap-filled daily SST analyses from NASA JPL and the Met Office.

NaN — “Not a number”, the stored value for “not measured”.

NASA — The US space agency.

NCAR — The US National Center for Atmospheric Research.

NCEI, OCADS — NOAA’s National Centers for Environmental Information, the US national environmental archive, and its ocean-carbon archive.

NCEP — The US National Centers for Environmental Prediction, producer of the reanalysis in family 7.2.

NDBC — NOAA’s National Data Buoy Center.

Niño-3.4, Niño indices — The average SST of fixed boxes in the equatorial Pacific; Niño-3.4 is the standard El Niño index.

NOAA, NOAA STAR — The US National Oceanic and Atmospheric Administration, and its satellite research centre.

NRT — Near-real-time: data released within days, before final checks.

NSIDC — The US National Snow and Ice Data Center.

NumPy, .npy — The standard Python array library and its array file format.

O₂, NO₃, PO₄, SiO₄, pH — Dissolved oxygen, nitrate, phosphate and silicate, and the acidity of seawater.

oc4k — Family 1.gf store of daily 4 km OC-CCI ocean colour.

OC-CCI — ESA’s Ocean Colour Climate Change Initiative record, merging several sensors since 1997.

OceanSITES — The network of open-ocean moored time-series stations.

oceansites — Family 1.gf store of OceanSITES open-ocean moorings, tropical arrays excluded.

OCO-2, OCO-3 — NASA’s Orbiting Carbon Observatories, which measure column CO₂ from a satellite (2014) and from the Space Station (2019).

OISST — NOAA’s daily Optimum Interpolation SST analysis at 0.25°.

OPeNDAP, THREDDS — Two server protocols for reading subsets of scientific files over HTTP.

OSI-408-a — OSI SAF’s sea-ice concentration product on a 10 km grid, computed from an 18 km footprint.

OSI SAF — EUMETSAT’s Ocean and Sea Ice Satellite Application Facility.

PACE, OCI — NASA’s Plankton, Aerosol, Cloud, ocean Ecosystem satellite (2024) and its hyperspectral Ocean Color Instrument.

Pathfinder — NOAA’s 4 km AVHRR SST record, 1981–2023, no longer updated.

phase A, B, C — The three build waves of each family: A cheap and first, B the terabyte stores, C decisions taken after A and B have verdicts.

PML — Plymouth Marine Laboratory, producer of OC-CCI.

precip01 — Family 1.gf store of daily 0.1° IMERG rainfall (exception E2).

pre-signed URL — A download link that works for a limited time without an account.

probe — A one-month (or one-zone) test run of a store’s real fetch code, measuring rows, bytes and speed before any full build.

public track, private track — Where a store lives: the public Hugging Face repository that anyone can read, or a private one used only by this project because the source’s licence forbids republishing.

QA, QC (qc) — Quality assurance and quality control: the producer’s flags saying how far a value can be trusted, kept as a channel or in the qc column.

QuikSCAT — NASA’s scatterometer satellite of 1999–2009.

read-out — A quantity the model is asked to predict or report, such as the Niño-3.4 index or next year’s forest loss.

reanalysis — A weather or ocean model run over past decades and continually corrected towards observations, giving complete fields that are partly model.

registry — The JSON file listing every store of a family with its tier, location, channels, checksums and licence; the one file a reader opens first.

RG (Roemmich–Gilson pressures) — The 16 standard depths (2.5 to 2,000 dbar) of the Roemmich–Gilson Argo climatology, used as the vertical axis of every profile store.

Rrs — Remote-sensing reflectance, the colour of light leaving the water at each wavelength.

RSS — Remote Sensing Systems, a company producing satellite climate records.

S3 (s3://) — Amazon’s object-storage service and its address form; many archives serve files through it.

scatterometer — A radar that infers wind speed and direction from the roughness of the sea surface.

schema 2, schema 3 — The column layout of a point store: schema 3 widens the time column to 64 bits so records before 1914 fit.

SeaFlux — A satellite-based estimate of the air–sea CO₂ flux.

seaice_asi — Family 1.gf store of daily 6.25 km sea-ice concentration from Bremen’s ASI algorithm.

seed — One independent training run; three seeds means the same test run three times with different random starts.

Sentinel-1, -2, -3, -5P, -6A (S1A, S1C, S1D, S2A, S2B, S2C) — The EU Copernicus satellites: C-band radar, 10 m optical, ocean colour and temperature, trace gases, and sea-level altimetry; the letters are individual satellites of a series.

shard, tile; shard_index.npy, tile_grid.json, n.tiles — In the G-fine layout, one compressed file per group and bin, cut into 256 × 256-pixel tiles; the index gives each tile’s byte position and the grid file each tile’s corners.

siblings — A list in family 10.2’s registry naming the registries of families 1.gf, 1.0.tf and 0.9.tf.

slatrack — Family 10.1’s store of satellite sea level along the ground track, all missions (67 GB).

SLSTR — The Sea and Land Surface Temperature Radiometer on Sentinel-3.

SMAP — NASA’s Soil Moisture Active Passive satellite, which also measures sea-surface salinity.

SMOS — ESA’s Soil Moisture and Ocean Salinity satellite.

SOCAT — The Surface Ocean CO₂ Atlas of ship measurements.

socat — Family 10.1’s store of SOCAT ship CO₂ measurements.

SSS — Sea-surface salinity.

SST — Sea-surface temperature.

sst_acspo02 — Family 1.gf store of daily 2 km SST merged across sensors by NOAA’s ACSPO.

sst_cci05 — Family 1.gf store of daily 5 km SST from ESA’s SST CCI, merged across sensors by the family’s own builder.

STAC — SpatioTemporal Asset Catalog, the standard in which producers list their scenes and files.

store — One dataset of a family, published as a small set of arrays plus a description file.

store.json — The description file of a store: its arrays, their checksums, channels and counts.

swath — The strip of ground a satellite instrument sees on one pass.

swh — Family 1.gf store of significant wave height along altimeter tracks.

swot — Family 1.gf store of SWOT’s 2 km swath sea-surface height, as point rows.

SWOT; LR, SSH — The Surface Water and Ocean Topography satellite (2022), which maps sea-surface height (SSH) across two wide swaths; LR is its low-rate ocean product.

TAO — The tropical Pacific mooring array, held in family 10.1’s **gtmba**.

tier G-fine (sharded) — The layout family 1.gf adds for grids finer than 0.25°: each five-day bin is one compressed file cut into tiles, so a read fetches only the tiles it needs.

tier G, P, T — Family 10’s three storage forms: G, gridded fields read one time step at a time; P, points, profiles and tracks read by nearest-neighbour search; T, image tiles held as a catalogue plus codec tokens.

token — The unit the model reads: a value, its offset in space and time from the anchor, and its footprint.

token_schema — The registry’s definition of the token’s fields.

Topex — The first precision radar altimeter satellite (1992), start of the sea-level record.

TROPOMI — The trace-gas imager on Sentinel-5P, measuring NO₂, CO and CH₄ (nitrogen dioxide, carbon monoxide, methane).

T, S, *u*, *v* — Temperature, salinity, and the east and north components of velocity or wind.

UK, US, EU — United Kingdom, United States, European Union.

URL — A web address.

verdict — Registry field holding a store’s falsifier result, empty until the test has run.

VIIRS; S-NPP, NOAA-20, NOAA-21 — The Visible Infrared Imaging Radiometer Suite, MODIS’s successor, on the Suomi-NPP (ending 2026-11), NOAA-20 and NOAA-21 satellites; I-band is its 375 m channel set.

wind12 — Family 1.gf store of 12.5 km scatterometer wind cells (exception E1).

WIND_GLO_PHY_L3_MY_012_005, WAVE_GLO_PHY_SWH_L3_MY_014_005, 009_103 — Copernicus Marine product identifiers: gridded scatterometer wind, along-track wave height, and GlobColour 300 m ocean colour.

wod — Family 1.gf store of World Ocean Database profiles other than Argo.

WOD; OSD, CTD, MBT, XBT, MRB, DRB, PFL, UOR, APB, GLD — The World Ocean Database, and its instrument classes: bottle casts, conductivity-temperature-depth casts, mechanical and expendable bathythermographs, moored and drifting buoys, profiling floats, undulating recorders, instrumented animals, gliders.

XCO₂ — The CO₂ concentration averaged through the whole depth of the atmosphere.

xco2 — Family 1.gf store of OCO-2, OCO-3 (and GOSAT) column-CO₂ soundings.

zstd — Zstandard, a fast lossless compressor.